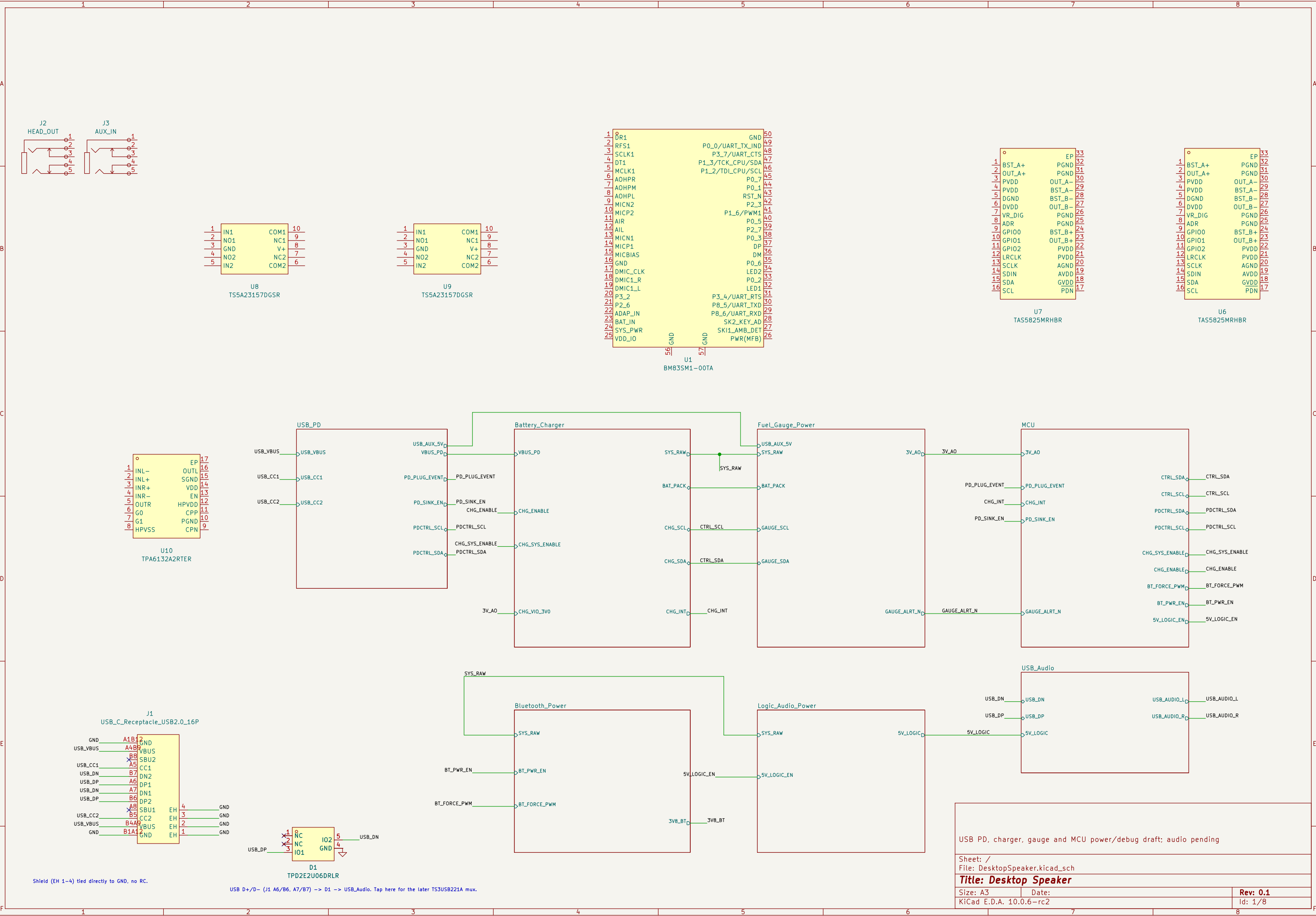
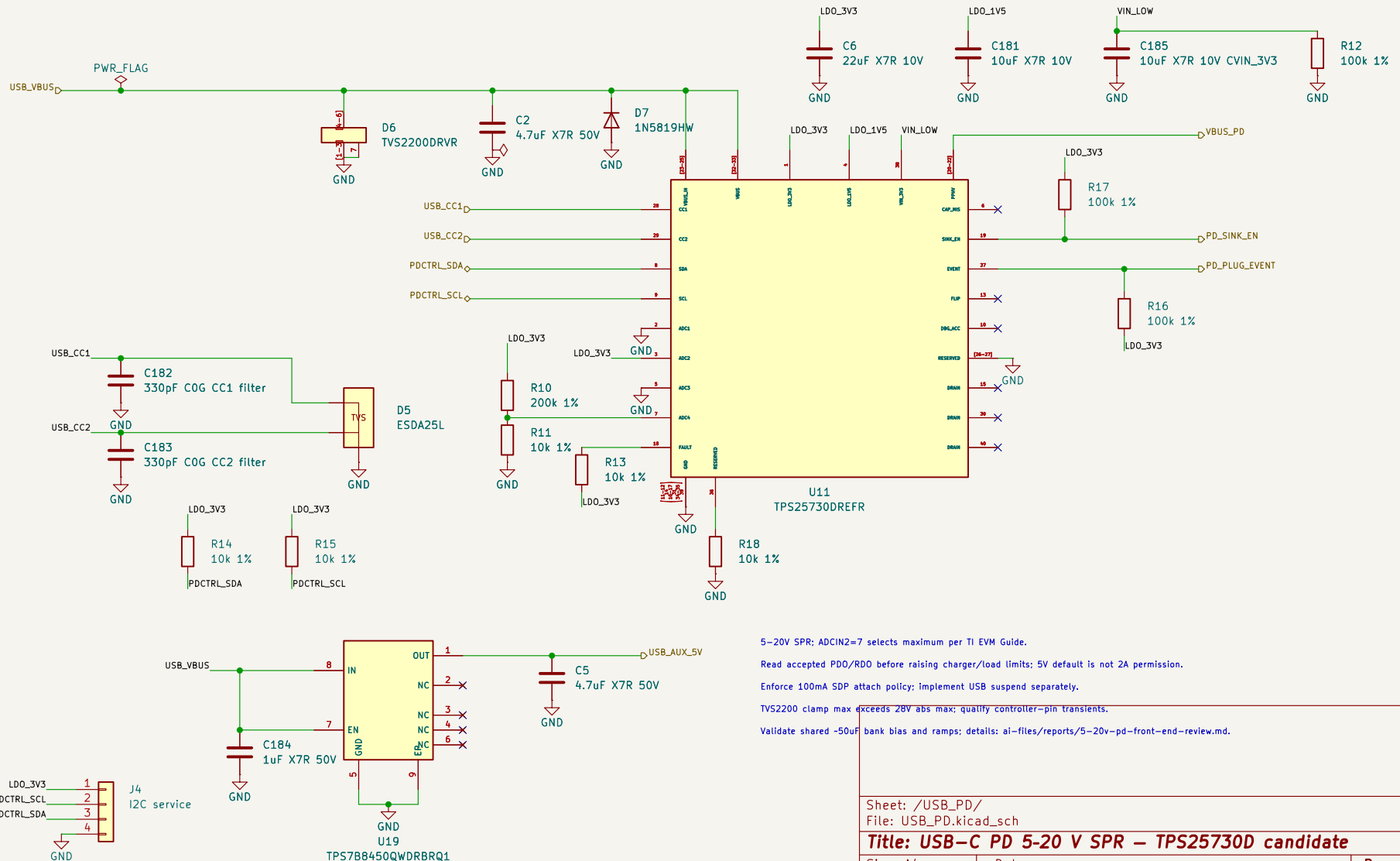
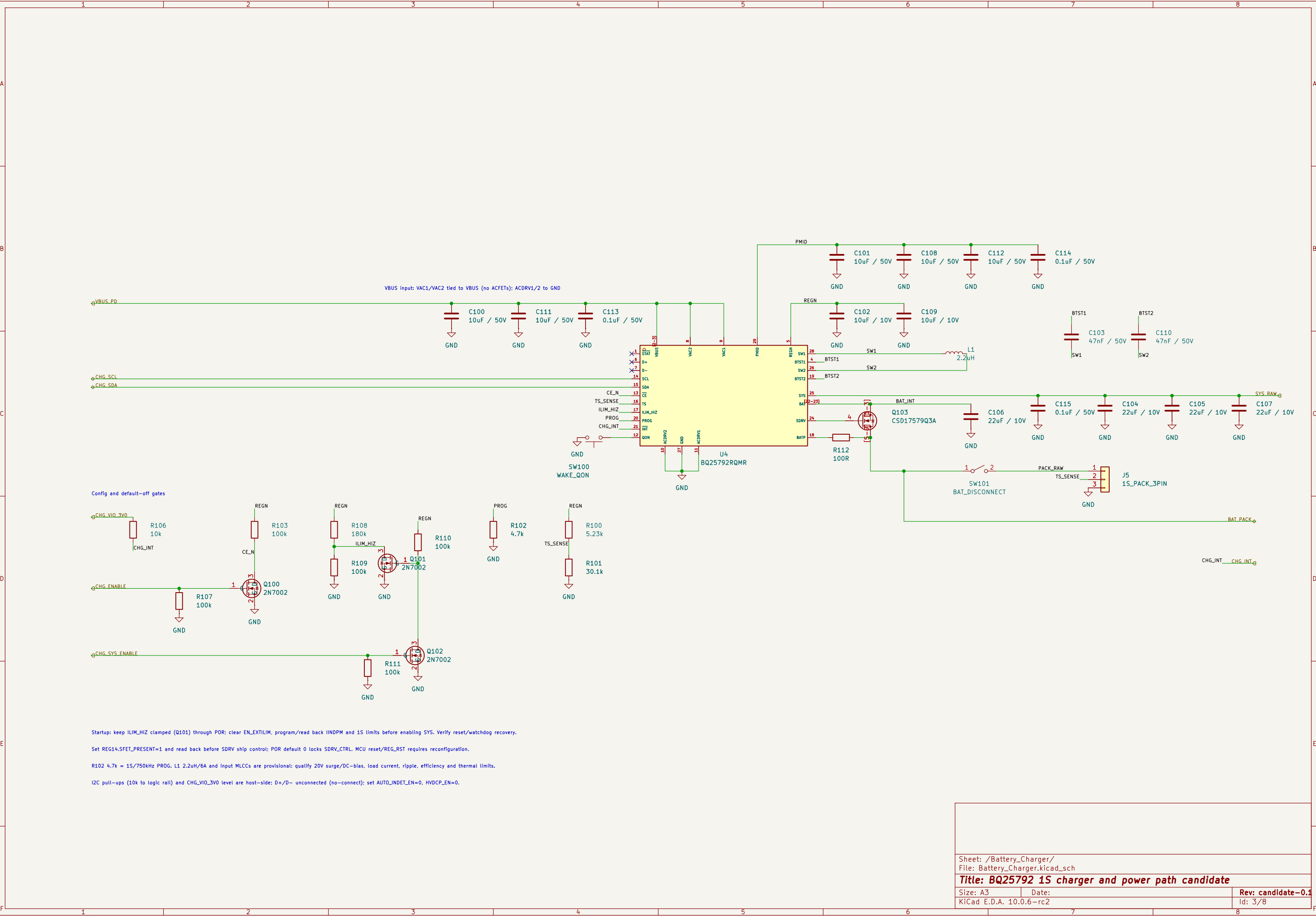
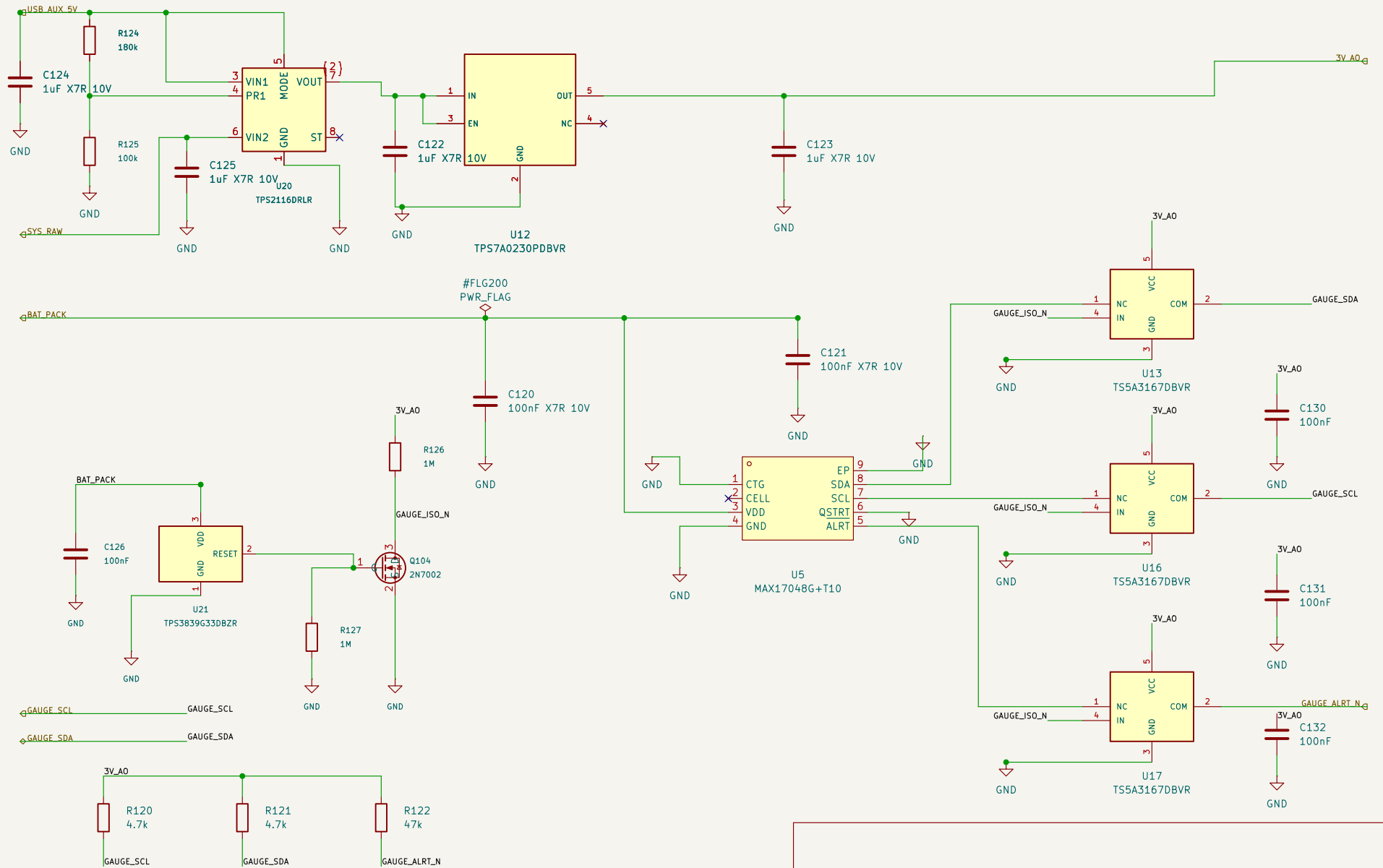










USB—priority logic supply; PR1 nominal 2.8 V.
Switchover/BOR and effective capacitance need qualification.

CELL pad 2 unused on MAX17048; QSTRT tied low

Gauge bus; AO—powered switches; default OFF below valid battery.
U21 3.08 V supervisor enables Q104 after reset recovery.

Sheet: /FuelGauge_Power/
File: FuelGauge_Power.kicad_sch

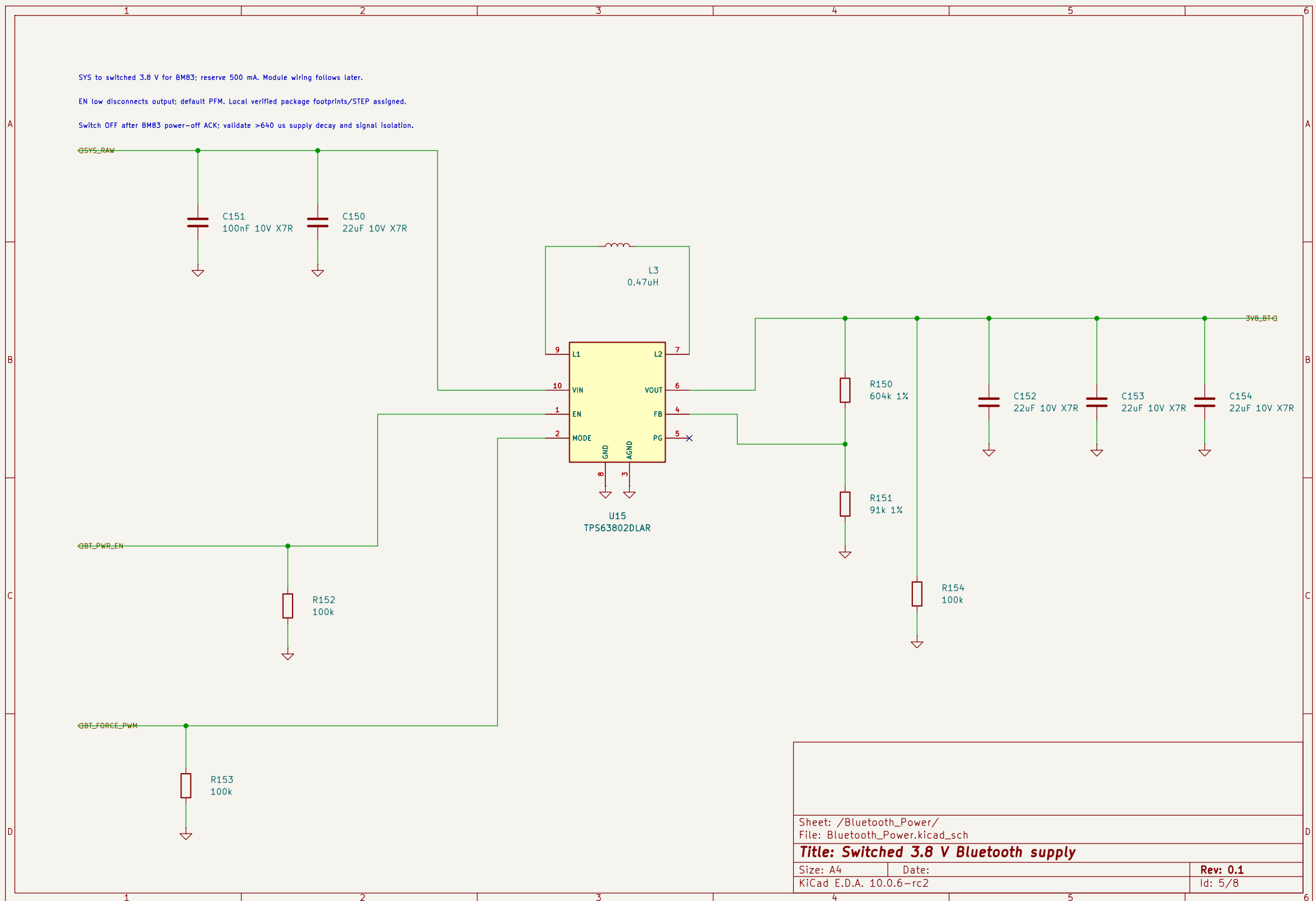
Title: Fuel gauge and low-current power

Size: A4
KiCad E.D.A. 10.0.6-rc2

Date:

Rev: 0.1

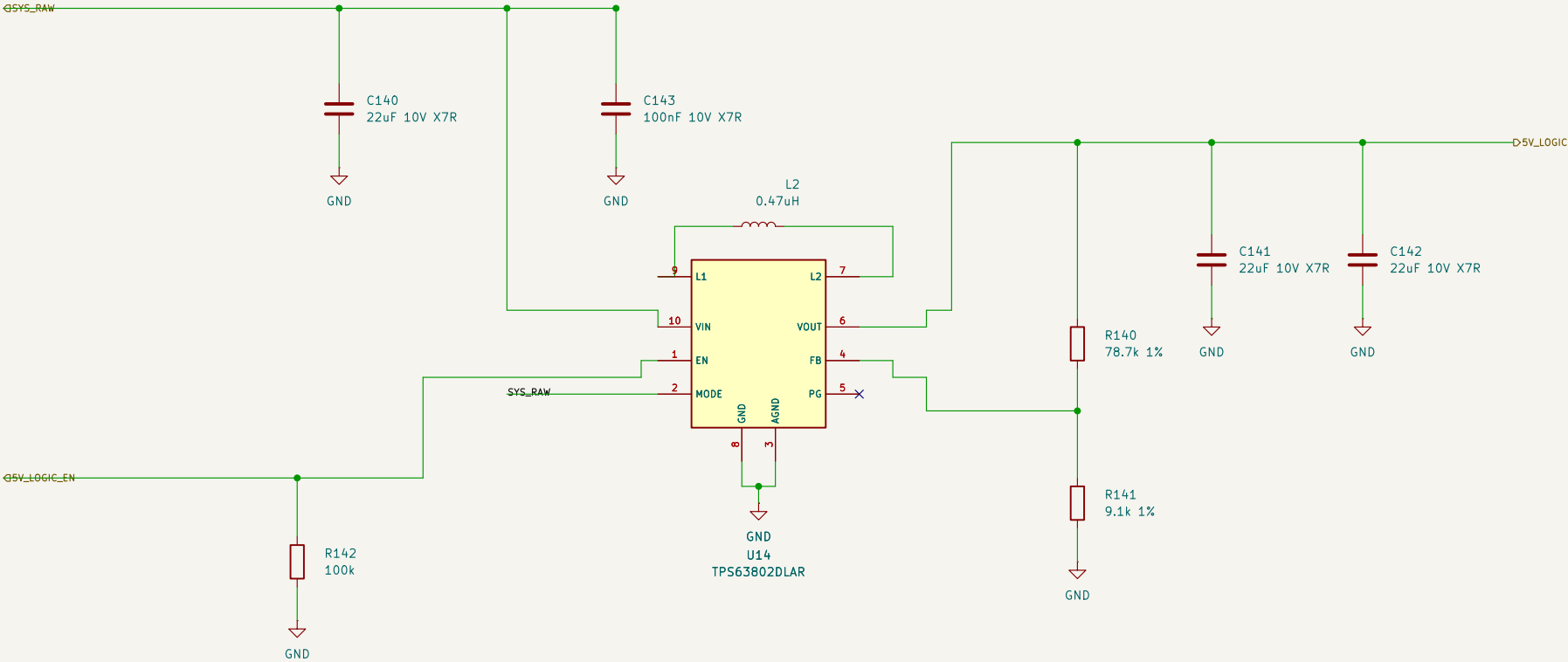
Id: 4/8



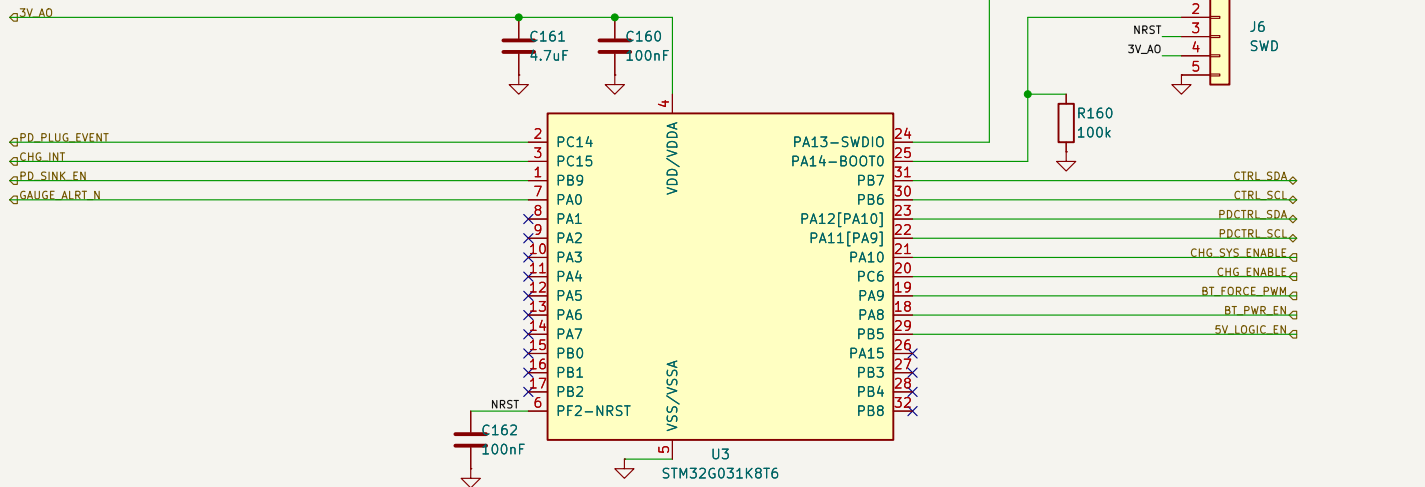
Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



3V_A0 supply: 100nF + 4.7uF at VDD/VDDA (DS12992 fig. 13); NRST 100nF, Internal pull-up



Reserved (no-connect): see ai-files/reports/mcu-pin-allocation.md

STM32G031K8T6: supply, reset/boot, SWD and connections to captured power blocks

Sheet: /MCU/
File: MCU.kicad_sch

Title: MCU and programming

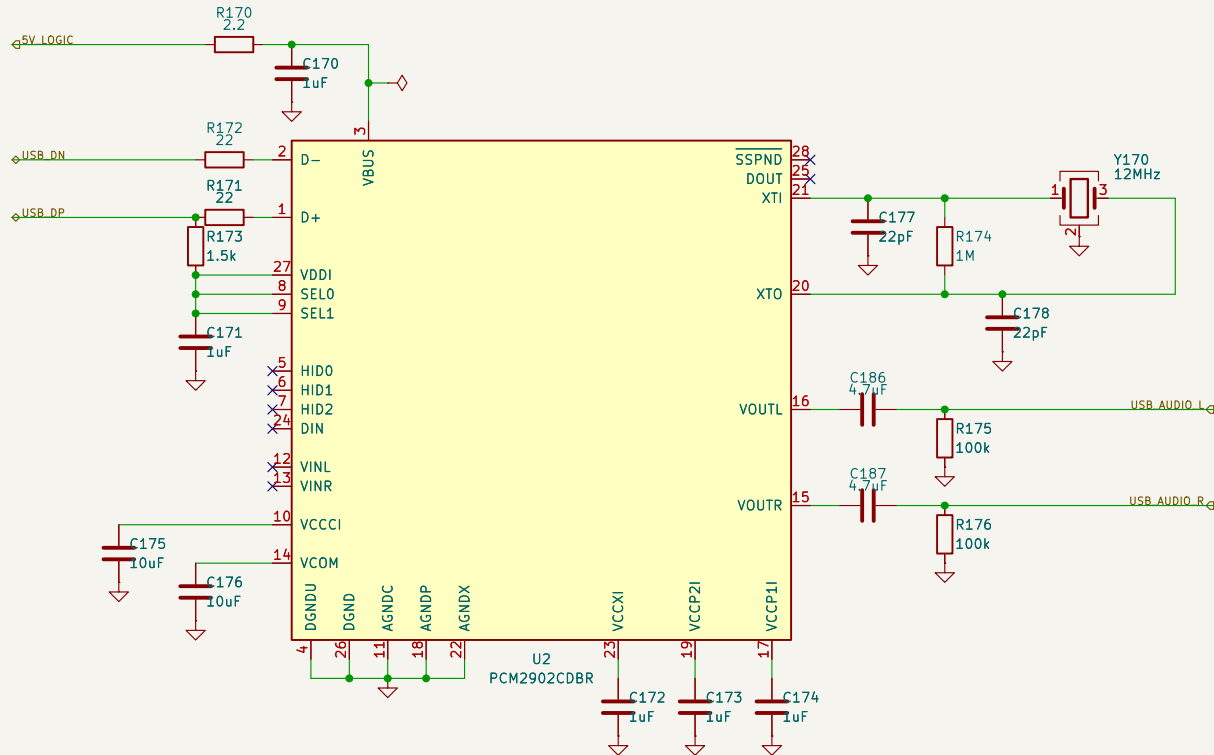
Size: A4
KiCad E.D.A. 10.0.6-rc2

Date:

Rev: 0.1

Id: 7/8

Supply: 5V_LOGIC (U14 TP563802, 4.60–5.06 V worst case) feeds VBUS through 2.2R/1uF. DS range 4.35–5.25 V, 56 mA typ / 67 mA max, 250 uA suspend.
 Codec is powered only when 5V_LOGIC_EN is high: D+ pull-up (R173 to VDDI) = USB attach. Do not enable on DCP/unknown sources.
 SEL0/SEL1 = high (VDDI). HID0–2, DIN, VINL/R unused. SSPND (suspend flag, active low) reserved for a later MCU input.
 USB_DP/USB_DN come from J1 via D1 (root). Later source-detect mux goes between those root nets and these ports.



Analogue out: 0.6 VCCCI Vpp (about 2.0 Vpp, 0.7 Vrms) centred on 1.65 V; AC-coupled (HPF -0.4 Hz at 100k), >=10k load.

PCM2902C bus-powered codec from 5V_LOGIC; USB D+/D- path; analogue out to USB_AUDIO_L/R

Sheet: /USB_Audio/
 File: USB_Audio.kicad_sch

Title: USB audio

Size: A4 Date: Rev: 0.1
 KiCad E.D.A. 10.0.6-rc2 Id: 8/8